Infinite Dyck Paths Algebra for Unbounded Interval Temporal Logic

Introduction

Three readings of a Dyck path

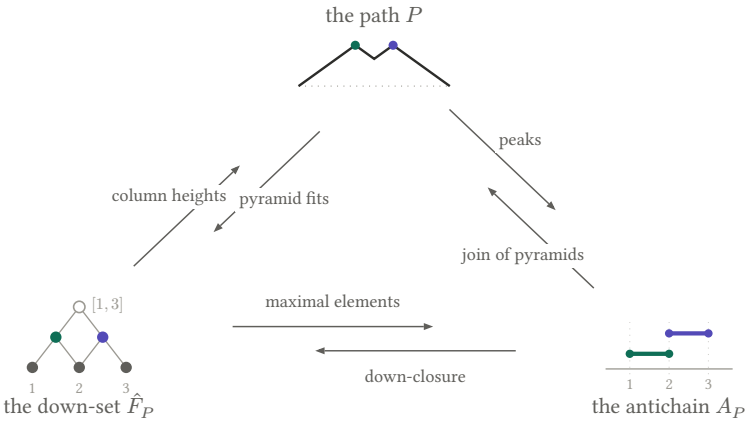
In this work we explore the duality between an interval temporal logic, Dyck paths, and prime filters. An interval temporal logic is characterized by the fact that the truth of a statement depends on the time interval it is evaluated on (rather than the time instant) [1]. Dyck paths enter as the geometric representation of such intervals sharing some algebraic properties with logic of subintervals [2]. The analysis of the finite Dyck lattices in [2] rests on one observation: a finite Dyck path P can be read in three ways.

- A path, an element of the lattice of Dyck paths ordered by geometric inclusion;
- a down-set named $\hat{F}_P$ of intervals of the underlying time line;
- an antichain of intervals written A_P, namely the maximal elements of that down-set.

These are three names for one object. Write D_n for the Dyck paths of semilength n ordered by geometric inclusion $P \leq Q$ when P lies weakly below Q [2, § 2]. This is a finite distributive lattice, so by Birkhoff's representation theorem [3] it is the lattice of down-sets of its join-irreducible elements, and those are transposed in Dyck algebra where join-irreducible is a path all of whose factors are hills except for a single pyramid $u^k d^k$ with $k \geq 2$, and it is determined by the interval of atoms lying below it. Writing $\mathrm{Int}([n-1])$ for the intervals $[i, j]$, $1 \leq i \leq j \leq n-1$, of the chain $[n-1]$ ordered by inclusion, this reads

$$D_n \cong \mathcal{O}(\mathrm{Int}([n-1])),$$

the lattice of down-sets of $\mathrm{Int}([n-1])$.



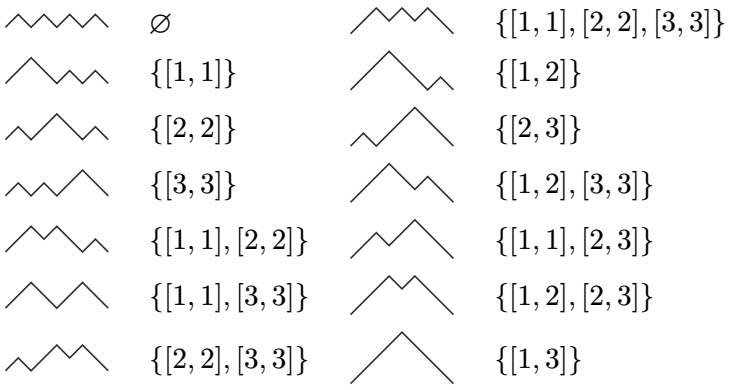
reading	$P = uuududdd$	read off from
the path P		the peaks above the axis at abscissas 3 and 5
the antichain A_P	$\{[1,2], [2,3]\}$	one maximal interval per peak of P above the axis
the down-set $\hat{F}_P$	$\mathrm{Int}([3]) \setminus \{[1,3]\}$	the down-closure of A_P

Figure 1: The three readings of $P = uuududdd \in D_4$.

Example. Take $n = 4$, so the time line is the chain $[3] = \{1, 2, 3\}$ and $\mathrm{Int}([3])$ has the six elements $[1,1], [2,2], [3,3], [1,2], [2,3], [1,3]$. For $P = uuududdd$, with height profile $0, 1, 2, 3, 2, 3, 2, 1, 0$, the test $h_P(i + j) \geq j - i + 2$ runs over those six:

$$\begin{array}{llllll} [1,1] & h_P(2) = 2 \geq 2 & [2,2] & h_P(4) = 2 \geq 2 & [3,3] & h_P(6) = 2 \geq 2 \\ [1,2] & h_P(3) = 3 \geq 3 & [2,3] & h_P(5) = 3 \geq 3 & [1,3] & h_P(4) = 2 \not\geq 4 \end{array}$$

So $\hat{F}_P$ is everything but $[1,3]$, and its maximal elements are $A_P = \{[1,2], [2,3]\}$ — the same answer the peak rule gives at a glance. Running the computation over all of D_4 labels each of its fourteen paths with an antichain of intervals of $[3]$:



Reading the two columns against Figure 4: the empty antichain is $\bot_4$, the single interval $[1,3]$ spanning the whole chain is $\top_4$, and the number of intervals in A_P is the number of nontrivial peaks of P.

From a bounded time line to an unbounded one

Everything above is finite with the chain $[n-1]$: the intervals are the time spans of a bounded history, and the lattice they generate is bounded with them. The suggestion this paper follows is to let that chain run off to the right and keep the triangle standing, or as much of it as will stand. An unbounded Dyck path is still to be read as a down-set of intervals of the time line $\mathbb{N}_{\geq 1}$, hence as a hereditary proposition about an unbounded future, and the question becomes which connectives such paths support once no last stage closes them off. The passage between a path and its down-set survives intact; the antichain of maximal intervals does not always, since a down-set of an infinite poset may have no maximal elements at all, and we will keep the intervals but not the antichain.

What holds the triangle up is a representation theorem, and it is worth saying on which scale ours sits. Birkhoff's theorem is a statement about finite distributive lattices. Its general form for bounded distributive lattices is Priestley duality, which trades the finite poset of join-irreducibles for an ordered topological space; beside it stands the topology-free representation of the down-set lattice $\mathcal{O}(P)$ of an arbitrary poset P by its completely join-irreducible elements, the principal down-sets [3]. We will not need the strength of Priestley duality and do not use it, but our work follows this spectrum: the stages D_n are Birkhoff's case, and the ambient object in which we place their limit remains the down-set lattice of a poset, the finite intervals of $\mathbb{N}_{\geq 1}$, where the correspondence between the poset and the completely join-irreducible elements — now the paths made of a single pyramid — survives unchanged.

> Instead of considering a finite total order, one can consider either an infinite (but discrete) total order or a more general partial order. In the former case, the combinatorial structure that is involved could be some notion of infinite Dyck path, that is, a Dyck path having infinite length. There are at least two different ways to formalize this suggestion: either working in the **set of all Dyck paths** (of any length), which is naturally partially ordered, and can be probably interpreted

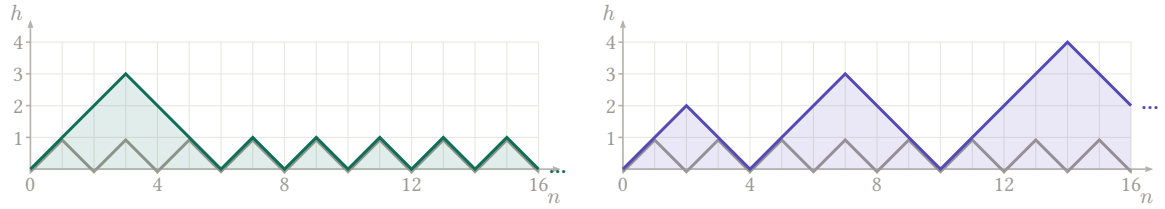
as some kind of infinitary limit object for the family of Dyck lattices; or introducing **Dyck paths having infinite support**, i.e., Dyck paths unbounded on the right which do not have a suffix consisting of the concatenation of an infinite number of peaks lying on the x-axis.

— Ferrari

Therefore, we denote:

- **Model 1:** the set of all Dyck paths;
- **Model 2:** the set of Dyck paths with infinite support.

Both models live inside the same picture. Read an infinite word over $\Sigma = \{u, d\}^\omega$ as a lattice path and compare it to the bottom element $\bot = (ud)^\omega$, the infinite concatenation of peaks lying on the axis: the two models are exactly the two ways a path can behave against $\bot$ in the long run.



Ambient definitions

For a path $P = p_0 p_1 p_2 \ldots$, we define the function $h_P(n)$ that returns the height of P after its first n steps.

$$h_P(0) = 0, \qquad h_P(n) = |\{i < n \mid p_i = u\}| - |\{i < n \mid p_i = d\}|.$$

$$\mathcal{D}_\omega = \{P = (p_i)_{i \in \mathbb{N}} \in \{u, d\}^\omega \mid \forall n \in \mathbb{N}, h_P(n) \geq 0\}$$

Only half of the finite condition survives here, and it is worth saying which half and why. A finite Dyck word is asked two things: that no prefix dip below the axis, and that the last step land back on it. The second speaks about the last step, and an infinite word has none, so $\mathcal{D}_\omega$ keeps the first alone. This is not an arbitrary weakening: it is what the finite notion becomes in the limit. A finite word w with no negative prefix is exactly a *prefix* of a finite Dyck word, since appending $d^{h_w(|w|)}$ closes it off; the definition above therefore reads

$$\mathcal{D}_\omega = \{P \in \{u, d\}^\omega \mid \forall n \in \mathbb{N}, \quad p_0 \cdots p_{n-1} \text{ is a prefix of a finite Dyck word}\},$$

the infinite branches of the tree of Dyck prefixes. Read the other way round, the paths of finite support — Model 1 seen inside $\mathcal{D}_\omega$ — are dense in it: truncating P at time n and closing it off with $d^{h_P(n)}(ud)^\omega$ gives a path of finite support agreeing with P up to n. Every infinite path is thus approximated by genuine finite Dyck paths, one prefix at a time, and nothing beyond that approximation has been let in.

Remark 1. An infinite Dyck path need not return to the axis. Returning is a feature of the finite case, forced there by the balance condition, and it lapses together with it: the descents a prefix owes are not missing from the path, only postponed, and postponement may go on forever. The extreme case is $\top = u^\omega$, which never comes back at all, although each of its prefixes u^n is settled by the honest Dyck path $u^n d^n$; a milder one is $u^k(du)^\omega$, which oscillates between heights $k - 1$ and k and never descends further.

Demanding infinitely many returns instead is possible, but costly. That set is not closed under joins — Proposition 21 writes u^ω as the supremum of the chain $u^k d^k (ud)^\omega$, every member of which returns infinitely often — so it is no complete lattice, and Model 2 would lose the top that is the only constant

of its logic. The return is a condition on the tail of a path, and the tail is exactly what an unbounded time line declines to fix.

Definition 2 (geometric order)**.** For $P, Q \in \mathcal{D}_\omega$ set

$$P \leq Q \quad \Leftrightarrow \quad \forall n \in \mathbb{N}, h_P(n) \leq h_Q(n),$$

and say that P lies *weakly below* Q. We call $\leq$ the *geometric order*, since it compares the two drawings pointwise. It restricts to each D_n, and $(D_n, \leq)$ is the Dyck lattice of order n of [2].

We import the vocabulary of [2] about specific Dyck paths, transposed to infinite words. A *return* of a path P is an index $n > 0$ with P crossing the x axis i.e. $h_P(n) = 0$; a *peak* is a pair of consecutive steps $p_n p_{n+1} = ud$, and a *hill* is a peak at height 0, i.e. one with $h_P(n) = 0$. A *pyramid* is a block $u^k d^k$ ($k \geq 1$) starting and ending on the axis, so a hill is the pyramid with $k = 1$. Finally, a *factor* of P is a block of consecutive steps running between two consecutive returns (or from the origin to the first return); a hill is also called a *trivial factor*. In the finite case every Dyck path is the juxtaposition of its factors; here the same holds for the prefix of P up to its last return, and a path with infinitely many returns is the juxtaposition of infinitely many factors; a path with no return at all, $\top = u^\omega$ being the extreme case, has no factors. The atom π_i are the path whose factors are all hills except for a single pyramid $uudd$.

Definition 3 (limit path)**.** In this language, we denote $\perp$ as the unique path all of whose factors are hills.

$$\perp = (ud)^\omega, \qquad h_\perp(n) = \begin{cases} 0 \text{ if } \ n = 0 \bmod 2 \\ 1 \text{ if } \ n = 1 \bmod 2 \end{cases}.$$

The *support* of a path measures how far it strays from the floor $\perp$: it collects the times at which P does not sit exactly where $\perp$ does. Since $\perp$ is nothing but hills, this amounts to recording where P does something other than a hill: a path of finite support is one whose factors are hills from some point on — a finite Dyck path followed by an infinite run of hills — whereas a path of infinite support keeps producing nontrivial factors arbitrarily late. This is precisely the line separating Model 1 from Model 2.

$$\mathrm{supp}(P) = \{n \in \mathbb{N} \mid h_P(n) \neq h_\perp(n)\}, \qquad \mathrm{supp}_{\mathrm{at}}(P) = \{i \geq 1 \mid h_P(2i) > 0\}.$$

The second variant samples only the even times, and there $\perp$ is on the axis: since paths never go below 0, "differing from $\perp$" and "being strictly above the axis" say the same thing. The next two lemmas show that this sampling loses nothing, so we may use whichever variant is convenient.

Lemma 4. $\mathrm{supp}_{\mathrm{at}}(P) = \{i \mid 2i \in \mathrm{supp}(P)\}$.

Lemma 5. $|\mathrm{supp}(P)| < \infty \Leftrightarrow |\mathrm{supp}_{\mathrm{at}}(P)| < \infty$.

Lemma 6. *$i \in \mathrm{supp}_{\mathrm{at}}(P) \Leftrightarrow \pi_i \leq P$, where $\pi_i = (ud)^{i-1}uudd(ud)^\omega$.*

Ferrari represents a path of D_n by an antichain of intervals of $[n-1]$, namely the orders of the atoms lying below each join-irreducible in its decomposition [2]. We keep the intervals, but not the antichain.

Definition 7 (interval representation)**.** Let Int denote the set of finite nonempty intervals $I = [a, b]$ of $\mathbb{N}_{\geq 1}$, ordered by inclusion. For $I = [a, b] \in \mathrm{Int}$ put

$$J_I = (ud)^{a-1} u^{b-a+2} d^{b-a+2} (ud)^\omega,$$

the path all of whose factors are hills except for a single pyramid, and set

$$\hat{F}_P = \{I \in \mathrm{Int} \mid J_I \leq P\}.$$

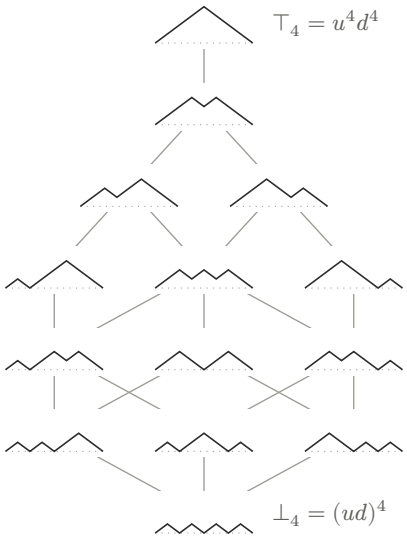
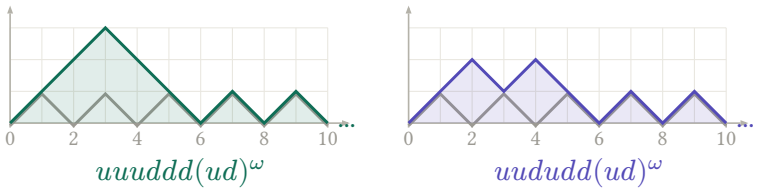


Figure 4: The Dyck lattice D_4, drawn in the geometric order.

Each J_I is join-irreducible in $\mathcal{D}_\omega$, and $J_{[i,i]} = \pi_i$; hence $\mathrm{supp}_{\mathrm{at}}(P) = \{i \mid [i,i] \in \hat{F}_P\}$: the support records exactly the singletons of $\hat{F}_P$, and forgets every wider interval.

Lemma 8. *$J_I \leq J_{I'}$ if and only if $I \subseteq I'$. In particular $\hat{F}_P$ is a down-set of* Int.

Example. Knowing $\mathrm{supp}_{\mathrm{at}}(P)$ is not enough to know P. Both paths below have $\mathrm{supp}_{\mathrm{at}} = \{1,2\}$, yet $uududd(ud)^\omega < uuuddd(ud)^\omega$.



The order is exactly what the picture shows: at every abscissa the right path sits weakly below the left one, and that pointwise comparison of heights is all that $\leq$ means.

The two are nevertheless different paths. The left one is a single pyramid, hence join-irreducible with atom-interval $[1,2]$, while the right one is $\pi_1 \vee \pi_2$; what separates them is the down-set, $\hat{F} = \{[1,1],[2,2],[1,2]\}$ against $\hat{F} = \{[1,1],[2,2]\}$, of which $\mathrm{supp}_{\mathrm{at}}(P)$ retains only the singletons. So $\mathrm{supp}_{\mathrm{at}}(P)$ tells us whether P has finite support, but not which path P is; only $\hat{F}_P$ does that.

Lemma 9. *Pointwise* min *and* sup *of height functions of paths in $\mathcal{D}_\omega$ are height functions of paths in $\mathcal{D}_\omega$; hence $(\mathcal{D}_\omega, \leq)$ is a complete lattice with $\bot = (ud)^\omega$, $\top = u^\omega$, and binary meets distribute over arbitrary joins, so $\mathcal{D}_\omega$ is a complete Heyting algebra.*

Dyck paths lattice

Each $(D_n, \leq)$ is a finite distributive lattice. The Figure 4 illustrate the lattice for $n = 4$. The bottom is the path of four hills $\bot_4$, the top the single pyramid $\top_4$, and an edge joins two paths when one covers the other in the geometric order.

These finite lattices are the stages of a single tower. To pass from one stage to the next we leave the path alone and append a hill at the right end by the inclusion function ι defined below. The Figure 5 draws the case ι_3.

$$\iota_n : D_n \to D_{n+1}, \qquad \iota_n(P) = Pud.$$

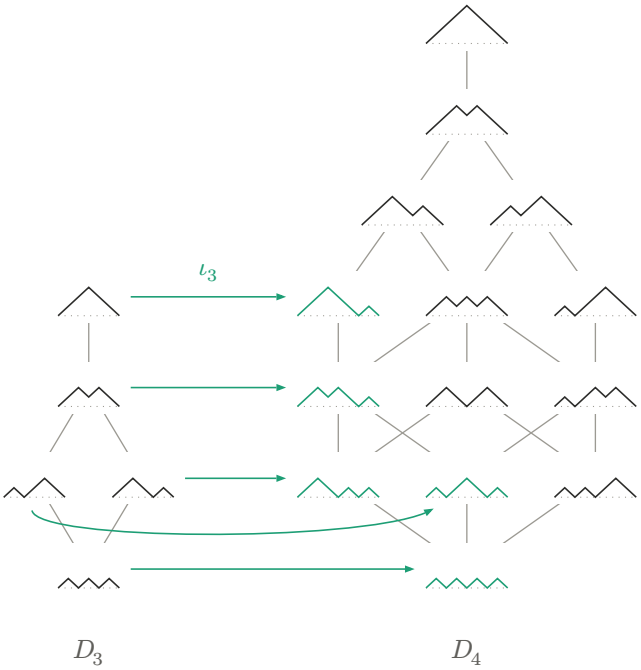


Figure 5: The case $\iota_3 : D_3 \hookrightarrow D_4$.

Appending a hill is also the extension that fixes the floor, sending $\bot_n = (ud)^n$ to $\bot_{n+1}$, which keeps the whole tower compatible with the reference path against which support is measured. In the interval representation the map does nothing at all: the intervals representing P are unchanged and only the ambient index set grows from $[n-1]$ to $[n]$, the new atom π_n failing to lie below Pud because Pud has returned to the axis at abscissa $2n$. Appending hills lets a path of one stage be compared with a path of any later one, and two paths should count as equal when the tower eventually makes them so.

Definition 10 (eventual equality)**.** For $n \le k$ let $\iota_{n,k} : D_n \to D_k$ denote the composite $\iota_{k-1} \circ \cdots \circ \iota_n$, with $\iota_{n,n} = \mathrm{id}$; that is, $\iota_{n,k}(P) = P(ud)^{k-n}$. On the disjoint union

$$\coprod_{n\in\mathbb{N}} D_n = \{(n, P) \mid n \in \mathbb{N}, P \in D_n\}$$

define

$$(n, P) \sim (m, Q) \quad \Leftrightarrow \quad \exists k \ge \max(n, m) : \iota_{n,k}(P) = \iota_{m,k}(Q).$$

The relation $\sim$ is obviously an equivalence relation. Its classes are what let paths of different lengths be compared at all: $P \in D_n$ and $Q \in D_m$ with $n \neq m$.

intro D_ω lattice and figure Figure 6

Model 1

We restate that Model 1 corresponds to the set of all Dyck paths (of any length), which is naturally partially ordered, and can probably be interpreted as some kind of infinitary limit object for the family of Dyck lattices [4], [5], [6].

Lemma 11. *ι_n is injective.*

Lemma 12. *ι_n is an order-embedding preserving function.*

The infinitary limit object $\mathcal{D}_{\text{colim}}$ for the family of Dyck lattices, denoted $\operatorname{colim} D_n$, is the colimit of the chain $D_0 \xrightarrow[\iota]{} D_1 \xrightarrow[\iota]{} D_2 \to \cdots$ along the maps ι_n: the smallest object in which every stage sits compatibly. Therefore, the $\mathcal{D}_{\text{colim}}$ is simply the increasing union of the tower quotiented by $\sim$.

Definition 13 (Direct limit of D_n)**.** Writing $[(n, P)]$ for the $\sim$-class of (n, P), put

$$\mathcal{D}_{\text{colim}} = \left(\coprod_{n \in \mathbb{N}} D_n \right) / \sim, \qquad c_n : D_n \to \mathcal{D}_{\text{colim}}, \quad c_n(P) = [(n, P)],$$

and equip $\mathcal{D}_{\text{colim}}$ with

$$[(n, P)] \leq [(m, Q)] \quad \Leftrightarrow \quad \exists k \geq \max(n, m) : \iota_{n,k}(P) \underset{D_k}{\leq} \iota_{m,k}(Q),$$

$$[(n, P)] \vee [(m, Q)] = \left[\left(k, \iota_{n,k}(P) \vee_{D_k} \iota_{m,k}(Q) \right) \right],$$

and

$$[(n, P)] \wedge [(m, Q)] = \left[\left(k, \iota_{n,k}(P) \wedge_{D_k} \iota_{m,k}(Q) \right) \right]$$

for any $k \geq \max(n, m)$, and $\bot = [(0, \varepsilon)]$ with ε the empty path.

An element of $\mathcal{D}_{\text{colim}}$ is thus a finite Dyck path together with the convention that the flat ground to its right does not count. Letting that ground run forever is exactly a subset of $\mathcal{D}_\omega$.

$$j : \mathcal{D}_{\text{colim}} \hookrightarrow \mathcal{D}_\omega \qquad j([(n, P)]) = P(ud)^\omega.$$

Proposition 14. *j is an order-embedding, and $j(\mathcal{D}_{\text{colim}}) = \{P \in \mathcal{D}_\omega \mid |\text{supp}(P)| < \infty\}$.*

Algebraic representation

Model 1 has a second possible description, it is an initial algebra. Recall that an algebra for an endofunctor F is just a map $F(X) \to X$ and a coalgebra is a map $X \to F(X)$; neither carries the associativity that an algebra over a monad, or a coalgebra over a comonad, would supply. Least and greatest fixpoints of F are written μF and νF respectively.

Definition 15 (initial algebra)**.** An *F-algebra* is a pair (X, α) with $\alpha : F(X) \to X$, and a morphism of F-algebras $(X, \alpha) \to (Y, \beta)$ is a map $f : X \to Y$ with $f \circ \alpha = \beta \circ F(f)$. An algebra $(\mu F, \text{in})$ is *initial* when for every F-algebra (X, α) there is exactly one such f, that is a unique

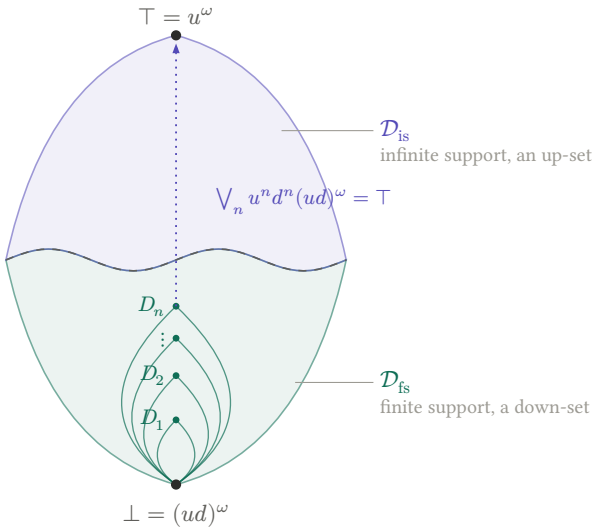


Figure 6: The two strata of $\mathcal{D}_\omega$, and the tower $D_1 \subset D_2 \subset \cdots D_\omega$.

$$\mathrm{fold}(\alpha) : \mu F \to X \quad \text{with} \quad \mathrm{fold}(\alpha) \circ \mathrm{in} = \alpha \circ F(\mathrm{fold}(\alpha)).$$

cite

Recall finite Dyck words are constructed with the grammar $S ::= \varepsilon \mid uSdS$ generates, and that grammar is a signature functor.

Definition 16 (Dyck initial algebra)**.** Let $F : \mathrm{Set} \to \mathrm{Set}$ be

$$F(X) = 1 + X \times X,$$

with constructors ε (i.e. $\varnothing$) for the terminal object 1 and $u(-)d(-)$ for the product $X \times X$, the latter reading a pair (A, B) as the word $uAdB$.

Proposition 17. *F is finitary, so its initial algebra exists and is computed by the initial chain $\varnothing \to F(\varnothing) \to F^2(\varnothing) \to \cdots$, which converges at ω:*

$$\mu F = \bigcup_{k \in \mathbb{N}} F^k(\varnothing),$$

with $\mathrm{in} : F(\mu F) \to \mu F$ *the pair of constructors. Its elements are the finite Dyck words before any quotient.*

Remark 18. The grammar is what makes $\top = u^\omega$ look illegal: no derivation from $S ::= \varepsilon \mid uSdS$ produces it. What is true is that no derivation *terminates* on it, and that is a statement about μ, not about the rules. Read as a least fixpoint the grammar gives μF, the words reached in finitely many productions, and u^ω is not one of them. Read as a greatest fixpoint it gives νF, the possibly infinite trees, where a derivation may run forever provided every single step is a legitimate production; and there u^ω is the word of the unique tree T with $T = uTdT$. Expanding the left argument forever emits u after u, and the d that each production promises sits behind an infinite word, so it is never reached: every rule is obeyed at every step, only the end of the derivation is missing. Flattening νF to words yields the finite Dyck words together with all of $\mathcal{D}_\omega$. The flattening is not injective — at a node whose left argument is infinite, the d and the right argument are unreachable and hence invisible in the word — which is why $\mathcal{D}_\omega$ is made of words and not of trees.

Let $\sigma : F(\mathbb{N}) \to \mathbb{N}$ be the F-algebra $\sigma(*) = 0$ and $\sigma(a, b) = a + b + 1$, and set

$$\ell := \mathrm{fold}(\sigma) : \mu F \to \mathbb{N}, \quad \ell(\varepsilon) = 0, \quad \ell(uAdB) = \ell(A) + \ell(B) + 1,$$

the unique map counting the uses of the constructor $u(-)d(-)$, i.e. the semilength.

Proposition 19. *The fibres of ℓ are the strata of the tower: $\ell^{-1}(n)$ is exactly the carrier of D_n, so that*

$$\mu F = \coprod_{n \in \mathbb{N}} \ell^{-1}(n) \cong \coprod_{n \in \mathbb{N}} D_n,$$

and $\left|\ell^{-1}(n)\right| = C_n$:

$$1, 1, 2, 5, 14, 42, 132, \ldots$$

the Catalan numbers.

define D_ω by the quotienting of μF

n	terms of μF	word	count
0	ε	ε	$C_0 = 1$
1	$u\,\varepsilon\,d\,\varepsilon$	ud	$C_1 = 1$
2	$u\,\varepsilon\,d\,(u\,\varepsilon\,d\,\varepsilon)$	$udud$	$C_2 = 2$
	$u\,(u\,\varepsilon\,d\,\varepsilon)\,d\,\varepsilon$	$uudd$	
3	$u\,\varepsilon\,d\,(u\,\varepsilon\,d\,(u\,\varepsilon\,d\,\varepsilon))$	$ududud$	$C_3 = 5$
	$u\,\varepsilon\,d\,(u\,(u\,\varepsilon\,d\,\varepsilon)\,d\,\varepsilon)$	$uduudd$	
	$u\,(u\,\varepsilon\,d\,\varepsilon)\,d\,(u\,\varepsilon\,d\,\varepsilon)$	$uuddud$	
	$u\,(u\,\varepsilon\,d\,(u\,\varepsilon\,d\,\varepsilon))\,d\,\varepsilon$	$uududd$	
	$u\,(u\,(u\,\varepsilon\,d\,\varepsilon)\,d\,\varepsilon)\,d\,\varepsilon$	$uuuddd$	

Table 1: The correspondance of μF and Catalan number $C_n = \frac{1}{n+1}\binom{2n}{n}$

Model 2

We restate that model 2 correspond to Dyck paths with infinite support, i.e., Dyck paths unbounded on the right which do not have a suffix consisting of the concatenation of an infinite number of peaks lying on the x-axis ($\ldots ud^\omega$).

$$\mathcal{D}_\infty = \{P \in \mathcal{D}_\omega \mid |\mathrm{supp}(P)| = \infty\}$$

Definition 20 (top path)**.**

$$\top = u^\omega, \qquad h_\top(n) = n, \qquad \mathrm{supp}(\top) = \mathbb{N}_{\geq 2}, \qquad \mathrm{supp}_{\mathrm{at}}(\top) = \mathbb{N}_{\geq 1}.$$

$$\top \in \mathcal{D}_\infty, \qquad \forall P \in \mathcal{D}_\infty, \quad P \leq \top.$$

After its first step $\top$ never meets the axis again: it has no return, hence no factor, and the grammar of finite Dyck words does not generate it. It is a Dyck path all the same, in the sense fixed by Remark 1 — every prefix u^n is a prefix of the Dyck path $u^n d^n$, and the descents it owes are postponed rather than dropped — and Remark 18 says the same on the side of the grammar, where $\top$ is the word of the tree $T = uTdT$ of νF.

Proposition 21. *$\top$ is the supremum of an increasing chain taken in stratum of Model 1:*

$$\top = \bigvee_{k\in\mathbb{N}} \underbrace{u^k d^k (ud)^\omega}_{\in \mathcal{D}_{\mathrm{fs}}} = \bigvee_{k\geq 2} \underbrace{u^k (du)^\omega}_{\in \mathcal{D}_\infty} = u^\omega$$

both because the k-th term has $h(n) = n$ as soon as $k \geq n$.

The left chain lies in Model 1 and its supremum does not, which is exactly why $\mathcal{D}_1$ has no top; the right chain shows that $\top$ is already a join of paths of infinite support, so Model 2 does not need Model 1 to name it.

Join

Proposition 22. *For $I \neq \varnothing$ and $P_i \in \mathcal{D}_\infty$,*

$$h_{\bigvee_{i\in I} P_i}(n) = \sup_{i\in I} h_{P_i}(n), \qquad \mathrm{supp}\left(\bigvee_{i\in I} P_i\right) = \bigcup_{i\in I} \mathrm{supp}(P_i),$$

hence $\bigvee_{i\in I} P_i \in \mathcal{D}_\infty$; in the binary case

$$h_{P\vee Q}(n) = \max\big(h_P(n), h_Q(n)\big), \qquad \operatorname{supp}(P \vee Q) = \operatorname{supp}(P) \cup \operatorname{supp}(Q).$$

$$\big(\mathcal{D}_\infty, \leq, \bigvee, \top\big) = \text{complete join-semilattice with top.}$$

Failure of meets

Lemma 23. $\operatorname{supp}_{\mathrm{at}}(P) = \varnothing \iff P = \bot$.

If $h_P(2i) = h_P(2i+2) = 0$ then $h_P(2i+1) = 1$, so $h_P = h_\bot$.

Example. Split the atoms π_i into the odd and the even ones:

$$O = (uudd)^\omega = \bigvee_{i \text{ odd}} \pi_i, \qquad E = ud(uudd)^\omega = \bigvee_{i \text{ even}} \pi_i,$$

$$\operatorname{supp}_{\mathrm{at}}(O) = \{1,3,5,\ldots\}, \qquad \operatorname{supp}_{\mathrm{at}}(E) = \{2,4,6,\ldots\}, \qquad O, E \in \mathcal{D}_\infty.$$

Their height functions interleave, and the pointwise minimum collapses onto the floor:

$$h_O = 0,1,2,1,0,1,2,1,0,\ldots$$
$$h_E = 0,1,0,1,2,1,0,1,2,\ldots$$
$$h_{O\wedge E} = 0,1,0,1,0,1,0,1,0,\ldots = h_\bot$$

$$\operatorname{supp}_{\mathrm{at}}(O \wedge E) = \operatorname{supp}_{\mathrm{at}}(O) \cap \operatorname{supp}_{\mathrm{at}}(E) = \varnothing \quad \Rightarrow \quad O \wedge E = \bot \notin \mathcal{D}_\infty.$$

Proposition 24. *O and E have no lower bound at all in $\mathcal{D}_\infty$:*

$$R \leq O, \quad R \leq E \quad \Rightarrow \quad R \leq \bot \quad \Rightarrow \quad R = \bot \notin \mathcal{D}_\infty.$$

Proposition 25. *$\wedge$ is a partial operation on $\mathcal{D}_\infty$:*

$$\operatorname{supp}(P \wedge Q) = \operatorname{supp}(P) \cap \operatorname{supp}(Q), \qquad P \wedge Q \in \mathcal{D}_\infty \iff |\operatorname{supp}(P) \cap \operatorname{supp}(Q)| = \infty.$$

$$\bot \notin \mathcal{D}_\infty, \qquad \mathcal{D}_\infty \text{ is not a meet-semilattice.}$$

Unbounded Interval Temporal Logics

$$\mathcal{D}_{\mathrm{fs}} := \{P \in \mathcal{D}_\omega \mid |\operatorname{supp}(P)| < \infty\}, \qquad \mathcal{D}_{\mathrm{is}} := \{P \in \mathcal{D}_\omega \mid |\operatorname{supp}(P)| = \infty\}.$$

$$\mathcal{D}_\omega = \mathcal{D}_{\mathrm{fs}} \sqcup \mathcal{D}_{\mathrm{is}}.$$

Logic of Model 1

$$\mathcal{D}_1 := \operatorname*{colim}_n D_n \cong \mathcal{D}_{\mathrm{fs}} = \{P \in \mathcal{D}_\omega \mid |\operatorname{supp}(P)| < \infty\}.$$

$$\bot = (ud)^\omega \in \mathcal{D}_1, \qquad \top \notin \mathcal{D}_1.$$

$$P, Q \in \mathcal{D}_1 \quad \Rightarrow \quad P \wedge Q \in \mathcal{D}_1, \qquad P \vee Q \in \mathcal{D}_1.$$

$$h_{P\wedge Q}(n) = \min\big(h_P(n), h_Q(n)\big), \qquad h_{P\vee Q}(n) = \max\big(h_P(n), h_Q(n)\big).$$

$$\operatorname{supp}(P \wedge Q) = \operatorname{supp}(P) \cap \operatorname{supp}(Q), \qquad \operatorname{supp}(P \vee Q) = \operatorname{supp}(P) \cup \operatorname{supp}(Q).$$

$$\mathcal{D}_1 = \text{distributive lattice with bottom, without top.}$$

$$\neg\exists\top \quad \Rightarrow \quad \neg\exists \text{ global Heyting implication.}$$

Nor does $\mathcal{D}_1$ keep the implication $\to$ and negation $\neg$ of $\mathcal{D}_\omega$ (Completion below): for $P, Q \in \mathcal{D}_1$, every J_I with I beyond $\mathrm{supp}(P)$ has $J_I \wedge P = \bot \leq Q$, hence $J_I \leq P \to Q$; so

$$P, Q \in \mathcal{D}_1 \quad \Rightarrow \quad |\mathrm{supp}(P \to Q)| = |\mathrm{supp}(\neg P)| = \infty, \qquad \neg\bot = \top \notin \mathcal{D}_1.$$

$$\mathcal{L}_1 = \{\bot, \wedge, \vee\}.$$

Logic of Model 2

$$\mathcal{D}_2 := \mathcal{D}_{\mathrm{is}} = \{P \in \mathcal{D}_\omega \mid |\mathrm{supp}(P)| = \infty\}.$$

$$(ud)^\omega \notin \mathcal{D}_2, \qquad \top = u^\omega \in \mathcal{D}_2.$$

$$P, Q \in \mathcal{D}_2 \quad \Rightarrow \quad P \vee Q \in \mathcal{D}_2.$$

$$\mathrm{supp}(P \vee Q) = \mathrm{supp}(P) \cup \mathrm{supp}(Q), \qquad |\mathrm{supp}(P)| = \infty \quad \Rightarrow \quad |\mathrm{supp}(P \vee Q)| = \infty.$$

$$P, Q \in \mathcal{D}_2 \quad \not\Rightarrow \quad P \wedge Q \in \mathcal{D}_2.$$

$$\mathrm{supp}(P \wedge Q) = \mathrm{supp}(P) \cap \mathrm{supp}(Q), \qquad |\mathrm{supp}(P)| = |\mathrm{supp}(Q)| = \infty \quad \not\Rightarrow \quad |\mathrm{supp}(P) \cap \mathrm{supp}(Q)| = \infty.$$

$$\bot \notin \mathcal{D}_2, \qquad \wedge \text{ is not closed.}$$

$$\mathcal{D}_2 = \text{join-semilattice with top}, \qquad \mathcal{L}_2 = \{\top, \vee, \to\}.$$

$$\neg\exists\bot, \qquad \neg\exists \text{ global } \wedge, \qquad \neg\exists \text{ global negation}: \quad \neg\top = \bot \notin \mathcal{D}_2.$$

The implication of $\mathcal{D}_\omega$, however, restricts to $\mathcal{D}_2$: $Q \leq P \to Q$, and $\mathcal{D}_2$ is an up-set of $\mathcal{D}_\omega$ since supp is monotone, so

$$P, Q \in \mathcal{D}_2 \quad \Rightarrow \quad P \to Q \in \mathcal{D}_2.$$

It is not the residual of the partial meet of $\mathcal{D}_2$: $O \to E = \neg O = E$, while no $R \in \mathcal{D}_2$ has $O \wedge R \in \mathcal{D}_2$ below E, as $O \wedge R \leq O \wedge E = \bot$. The adjunction $P \wedge R \leq Q \iff R \leq P \to Q$ is available in $\mathcal{D}_\omega$ only.

Completion

$$\mathcal{D}_\omega = \mathcal{D}_{\mathrm{fs}} \sqcup \mathcal{D}_{\mathrm{is}}, \qquad \mathcal{D}_1 \to \mathcal{D}_\omega \leftarrow \mathcal{D}_2.$$

$$\mathcal{D}_1 \cong \mathcal{D}_{\mathrm{fs}} \subseteq \mathcal{D}_\omega, \qquad \mathcal{D}_2 = \mathcal{D}_{\mathrm{is}} \subseteq \mathcal{D}_\omega.$$

$$\bot = (ud)^\omega, \qquad \top = u^\omega.$$

$$h_{\sup_{i\in I} P_i}(n) = \sup_{i\in I} h_{P_i}(n), \qquad h_{\inf_{i\in I} P_i}(n) = \inf_{i\in I} h_{P_i}(n).$$

$$P \wedge \left(\sup_{i\in I} Q_i\right) = \sup_{i\in I}(P \wedge Q_i).$$

$$\mathcal{D}_\omega = \text{frame} = \text{complete Heyting algebra}.$$

$$P \to Q = \sup\{R \in \mathcal{D}_\omega \mid P \wedge R \leq Q\}, \qquad \neg P = P \to \bot.$$

$$\mathcal{L}_\omega = \{\bot, \top, \wedge, \vee, \to, \neg\} + \{\sup_{i\in I}, \inf_{i\in I}\}.$$

Remark 26. The modalities of [2] are not missing from $\mathcal{L}_\omega$. Read on sets of intervals, $\Box U = \{I \mid \forall J \subseteq I, J \in U\}$ is the largest down-set inside U and $\Diamond U = \{I \mid \exists J \subseteq I, J \in U\}$ the smallest up-

set containing it; both are idempotent. On a down-set — a path — $\Box$ is the identity, so Θ_n is exactly the $\Box$-fixed fragment of ITL_n, and $\Box$ adds no connective to any of $\mathcal{L}_1$, $\mathcal{L}_2$, $\mathcal{L}_\omega$. $\Diamond$ does not preserve down-sets: $\Diamond\varepsilon_1$ holds on $[t_1, t_2]$ but not on $[t_2, t_2]$, so it has no reading as a path and belongs to none of them. The pseudo-connectives $\sim\varphi = \Box\neg\varphi$ and $\varphi \rightsquigarrow \psi = \Box(\varphi \to \psi)$ of [2] are the negation and implication of the frame, $\neg P = P \to \bot$: $\mathcal{D}_1$ keeps neither, $\mathcal{D}_2$ keeps $\to$ but not $\neg$.

$$\mathrm{Idl}(\mathcal{D}_1) \cong \mathcal{D}_\omega.$$

Final overview

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